

VERILOG מעבדה 6

מגישים:

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C.1- Test Vectors :

1.

1	8000000000000000	0000000000000000	95A8D72813DAA94D
2	4000000000000000	0000000000000000	0EEC1487DD8C26D5
3	2000000000000000	0000000000000000	7AD16FFB79C45926
4	1000000000000000	0000000000000000	D3746294CA6A6CF3
5	0800000000000000	0000000000000000	809F5F873C1FD761
6	0400000000000000	0000000000000000	C02FAFFEC989D1FC
7	0200000000000000	0000000000000000	4615AA1D33E72F10
8	0100000000000000	0000000000000000	8CA64DE9C1B123A7
9	0080000000000000	0000000000000000	2055123350C00858
10	0040000000000000	0000000000000000	DF3B99D6577397C8
11			

2.

```
# time = 170000 | K = 8000000000000000 | P = 0000000000000000 | C = 95a8d72813daa94d
# time = 250000 | K = 4000000000000000 | P = 0000000000000000 | C = 0eec1487dd8c26d5
# time = 330000 | K = 2000000000000000 | P = 0000000000000000 | C = 7ad16ffb79c45926
# time = 410000 | K = 1000000000000000 | P = 0000000000000000 | C = d3746294ca6a6cf3
# time = 490000 | K = 0800000000000000 | P = 0000000000000000 | C = 809f5f873c1fd761
# time = 570000 | K = 0400000000000000 | P = 0000000000000000 | C = c02faffec989d1fc
# time = 650000 | K = 0200000000000000 | P = 0000000000000000 | C = 4615aaid33e72f10
# time = 730000 | K = 0100000000000000 | P = 0000000000000000 | C = 8ca64de9c1b123a7
# time = 810000 | K = 0080000000000000 | P = 0000000000000000 | C = 2055123350c00858
# time = 890000 | K = 0040000000000000 | P = 0000000000000000 | C = df3b99d6577397c8
# ** Error: (vsim-PLI-3485) tf_getp(): Unsupported type conversion. Use tf_strgetp().
# Time: 970 ns Iteration: 1 Instance: /tb_test_vectors/DUT
# 1
# Break in Module stub at Z:/verilog/lab_6/stub.v line 60
```

3.

```
1 95A8D72813DAA94D
2 0EEC1487DD8C26D5
3 7AD16FFB79C45926
4 D3746294CA6A6CF3
5 809F5F873C1FD761
6 C02FAFFEC989D1FC
7 4615AA1D33E72F10
8 8CA64DE9C1B123A7
9 2055123350C00858
10 DF3B99D6577397C8
```

c.2- DES Stub and self comparing Testbench:

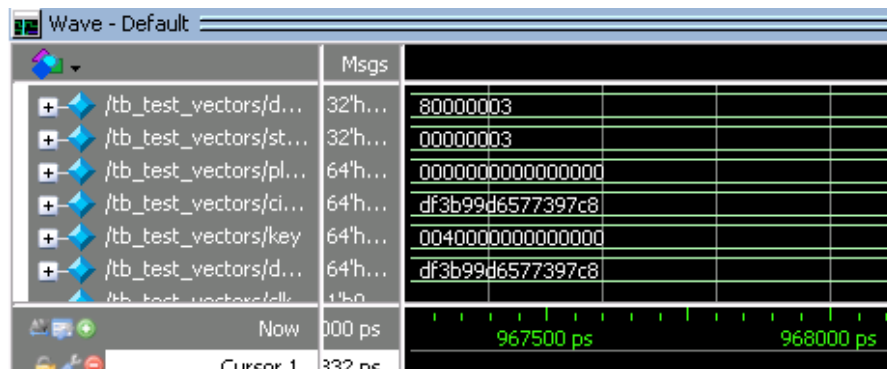
1. The stub code:

```
1 `timescale 1 ns / 1 ps
2 module stub
3 (
4     input clk ,
5     input reset ,
6     input start ,
7     input [63:0] data_in ,
8     input [63:0] key ,
9     output reg [63:0] data_out,
10    output reg ready
11 );
12
13 localparam IDLE = 0;
14 localparam IS_READY = 1;
15 integer data_file_in;
16 integer statusD;
17 reg state;
18
19 initial
20     data_file_in = $fopen("test_vectors_chipher_only.txt", "r");
21
22 always @(posedge clk or posedge reset)
23
24 always @(posedge clk or posedge reset)
25     if (reset)
26     begin
27         ready <= 0;
28         data_out <= 0;
29     end
30     else
31     if (start == 1)
32     begin
33         ready <= 0;
34     end
35     else
36     if (state == IS_READY)
37     if ( ! $feof(data_file_in))
38     begin
39         statusD = $fscanf(data_file_in,"%h\n", data_out);
40         ready <= 1;
41     end
42     else
43     begin
44         $fclose(data_file_in);
45         ready <= 0;
46         $finish("stub_file is finished");
47     end
48 endmodule
```

2. Testbench:

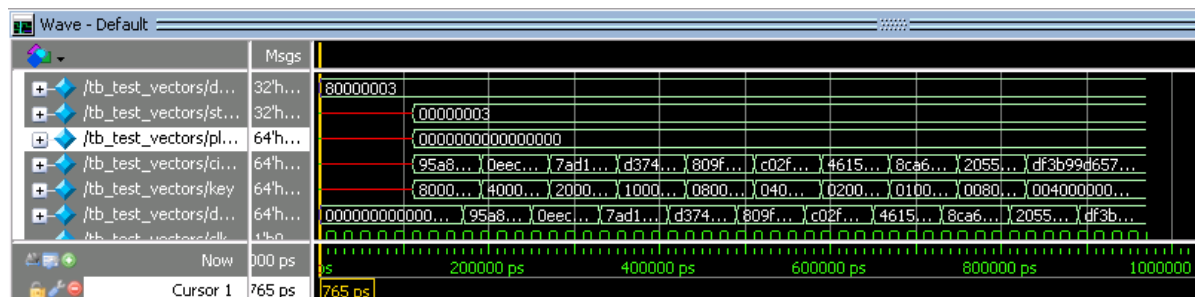
```
1  `timescale 1 ns / 1 ps
2
3  module tb_test_vectors();
4
5      integer    data_file_in;
6      integer    statusD;
7      reg  [63:0] plaintext, ciphertext, key ;
8      wire [63:0] data_out;
9      reg        clk = 0;
10     reg        reset;
11     reg        start = 0;
12     wire       ready;
13     reg  [1:0]  q;
14
15
16     always
17         #10 clk = ~ clk;
18
19     initial
20     begin
21         reset = 1;
22         #56 reset = 0;
23     end
24
25     initial
26         data_file_in = $fopen("test_vectors_first10.txt", "r");
27
28     always @ (posedge clk)
29         if ((q == 2) && ( ! $feof(data_file_in)))
30         begin
31             statusD = $fscanf(data_file_in,"%h %h %h\n" , key, plaintext, ciphertext);
32         end
33
34     always @ (posedge ready)
35         if (data_out == ciphertext)
36             $display("time = %8t | K = %016h | P = %016h | C = %016h | Result = %016h | PASS",
37                 $time, key, plaintext, ciphertext, data_out);
38         else
39             $display("time = %8t | K = %016h | P = %016h | C = %016h | Result = %016h | FAIL",
40                 $time, key, plaintext, ciphertext, data_out);
41
42     always @ (posedge clk or posedge reset)
43         if (reset == 1)
44         begin
45             q      <= 0;
46             start  <= 0;
47         end
48         else
49             if (q == 3)
50             begin
51                 q      <= 0;
52                 start  <= 1;
53             end
54             else
55             begin
56                 q      <= q + 1;
57                 start  <= 0;
58             end
59
60     stub DUT
61     (
62         .clk      (clk ),
63         .reset     (reset ),
64         .start     (start ),
65         .key       (key),
66         .data_in   (plaintext),
67         .data_out  (data_out),
68         .ready     (ready)
69     );
70
71 endmodule
```

3. log file PASS :



```
# time = 170000 | K = 8000000000000000 | P = 0000000000000000 | C = 95a8d72813daa94d | Result = 95a8d72813daa94d | PASS
# time = 250000 | K = 4000000000000000 | P = 0000000000000000 | C = 0eec1487dd8c26d5 | Result = 0eec1487dd8c26d5 | PASS
# time = 330000 | K = 2000000000000000 | P = 0000000000000000 | C = 7ad16fffb79c45926 | Result = 7ad16fffb79c45926 | PASS
# time = 410000 | K = 1000000000000000 | P = 0000000000000000 | C = d3746294ca6a6cf3 | Result = d3746294ca6a6cf3 | PASS
# time = 490000 | K = 0800000000000000 | P = 0000000000000000 | C = 809f5f873c1fd761 | Result = 809f5f873c1fd761 | PASS
# time = 570000 | K = 0400000000000000 | P = 0000000000000000 | C = c02faffec989d1fc | Result = c02faffec989d1fc | PASS
# time = 650000 | K = 0200000000000000 | P = 0000000000000000 | C = 4615aald33e72f10 | Result = 4615aald33e72f10 | PASS
# time = 730000 | K = 0100000000000000 | P = 0000000000000000 | C = 8ca64de9c1b123a7 | Result = 8ca64de9c1b123a7 | PASS
# time = 810000 | K = 0080000000000000 | P = 0000000000000000 | C = 2055123350c00858 | Result = 2055123350c00858 | PASS
# time = 890000 | K = 0040000000000000 | P = 0000000000000000 | C = df3b99d6577397c8 | Result = df3b99d6577397c8 | PASS
# ** Error: (vsim-PLI-3485) tf_getp(): Unsupported type conversion. Use tf_strgetp(). : Z:/verilog/lab_6/stub.v(60)
# Time: 970 ns Iteration: 1 Instance: /tb_test_vectors/DUT
```

4. log file FAIL:



```
# time = 170000 | K = 8000000000000000 | P = 0000000000000000 | C = 95a8d72813daa94d | Result = 95a8d72813daa94d | PASS
# time = 250000 | K = 4000000000000000 | P = 0000000000000000 | C = 0eec1487dd8c26d5 | Result = 0eec1487dd8c36d5 | FAIL
# time = 330000 | K = 2000000000000000 | P = 0000000000000000 | C = 7ad16fffb79c45926 | Result = 7ad16fffb79c45926 | PASS
# time = 410000 | K = 1000000000000000 | P = 0000000000000000 | C = d3746294ca6a6cf3 | Result = d3746294ca6a6cf3 | PASS
# time = 490000 | K = 0800000000000000 | P = 0000000000000000 | C = 809f5f873c1fd761 | Result = 809f5f873c1fd761 | PASS
# time = 570000 | K = 0400000000000000 | P = 0000000000000000 | C = c02faffec989d1fc | Result = c02faffec989d2fc | FAIL
# time = 650000 | K = 0200000000000000 | P = 0000000000000000 | C = 4615aald33e72f10 | Result = 4615aald33e72f10 | PASS
# time = 730000 | K = 0100000000000000 | P = 0000000000000000 | C = 8ca64de9c1b123a7 | Result = 8ca64de9c1b123a7 | PASS
# time = 810000 | K = 0080000000000000 | P = 0000000000000000 | C = 2055123350c00858 | Result = 2055123350c10858 | FAIL
# time = 890000 | K = 0040000000000000 | P = 0000000000000000 | C = df3b99d6577397c8 | Result = df3b99d6577397c8 | PASS
```

